

Class Overview

In Class:

- Genetic Algorithms & Testing
- Exploratory Testing
- Point “Shopping”
- (Optional) Final Exam Review

After Class Office Hours: Homework Help

Homework (Currently Assigned):

- 04-15: Testing with LLMs
- 04-22: Half-Semester Project

Clicker Points (Current): 6

Note: High volume of early reviews coming in!

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Tentative Remaining Schedule

Week	Plan
04/15 (Today)	Search-Based Testing, Exploratory Testing, Final Exam Review
04/22*	Final Assessment, Half-Semester Project Due
04/29	Half-Semester Project Presentations

*** No extensions past this point**

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Course Eval & Extra Credit

- Available on Friday, April 10 through Friday, April 24.
- If the following percent of people complete the survey:
 - **80%**, then **3pt** (out of 250) is added to the $\frac{1}{2}$ semester project.
 - **90%**, then **4pt** (out of 250) is added to the $\frac{1}{2}$ semester project.
 - **100%**, then **5pt** (out of 250) is added to the $\frac{1}{2}$ semester project.

Things I really want to hear about:

- Should the book be required?
- Are the early reviews helpful? Extensions?
- Should the clickers be used more or less?

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Course Eval & Extra Credit

Currently: 67%

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CIS 613: Genetic Algorithms & Testing

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Motivation

Given all these metrics, can we tell if a test case adds value?

Could we try random tests cases (i.e., inputs and outputs)
to see if they add value?

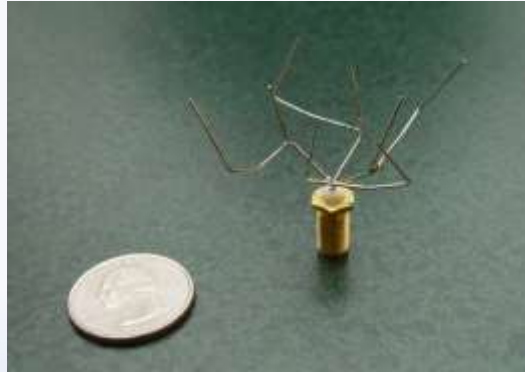
Could we do better than random search?

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Genetic Algorithm Applications: NASA ST5 Antenna

Evolved Antenna:

- Needs less power,
- More uniform coverage,
- Is more reliable,
- Requires less hardware,
- Took *less* time to develop:
 - 3 Person-Months vs
 - 5 Person-Months



Hornby, Gregory, et al. "Automated antenna design with evolutionary algorithms." *Space* 2006.

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Genetic Algorithm

- Directed search algorithms based on the mechanics of biological evolution
- Developed by John Holland, University of Michigan (1970's)
 - To understand the adaptive processes of natural systems
 - To design artificial systems software that retains the robustness of natural systems
- Provide efficient, effective techniques for optimization and machine learning applications
- Widely-used today in business, science and engineering

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What *is* a Genetic Algorithm

- Basic idea: Simulate natural selection, where the population is composed of candidate solutions.
- Genetic algorithms are a randomized heuristic search strategy.
- Focus is on evolving a population from which fit and diverse candidates can emerge via mutation and crossover.
- Genetic Operators:
 - Mutation emphasizes diversity
 - Crossover emphasizes diversity *and* fitness
 - Selection emphasizes fitness

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Components of a Genetic Algorithm

A problem to solve, and ...

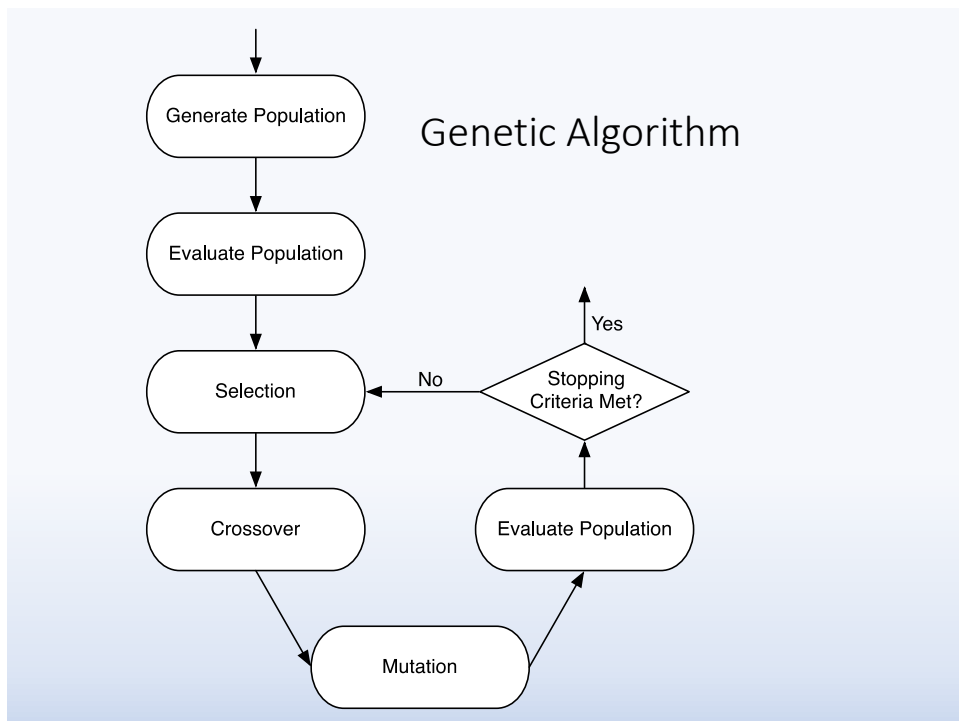
- Encoding technique: gene, chromosome
- Initialization procedure: creation
- Evaluation function: environment
- Selection of parents: reproduction
- Genetic operators: mutation, recombination
- Parameter settings: practice and art

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Genetic Algorithm

1. **Generate a population** of candidate solutions
2. **Evaluate:** Calculate the “fitness” of each member of the population
3. **Selection:** Probabilistically choose a subset of the population to survive
4. **Variation:** Permute individuals using genetic operators (crossover, mutation)
5. **Update Population:** Form next generation using selection and variation.
6. **Termination:** Stop when maximum fitness is achieved, or a specified number of generations.

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Genetic Algorithm: Population

Generate Population

Chromosomes could be:

- Bit strings: 0101 ... 1100
- Real numbers: 43.2 -33.1 ... 0.0 89.2)
- Permutations of element: E11 E3 E7 ... E1 E15
- Lists of rules: R1 R2 R3 ... R22 R23
- Program elements: genetic programming
- Or *any* other data structure

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Genetic Algorithm: Reproduction

Evaluate Population

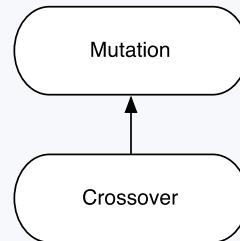


Selection

Parents are selected at random with selection chances biased in relation to chromosome evaluations.

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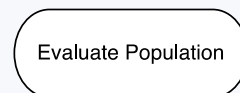
Genetic Algorithm: Modification



- Modifications are stochastically triggered
- Operator types are:
 - Mutation
 - Crossover (recombination)

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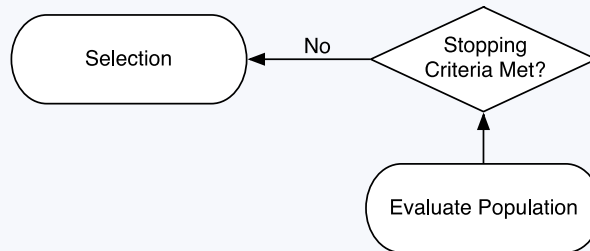
Genetic Algorithm: Evaluation



- The evaluator decodes a chromosome and assigns it a fitness measure
- The evaluator is the only link between a classical GA and the problem it is solving

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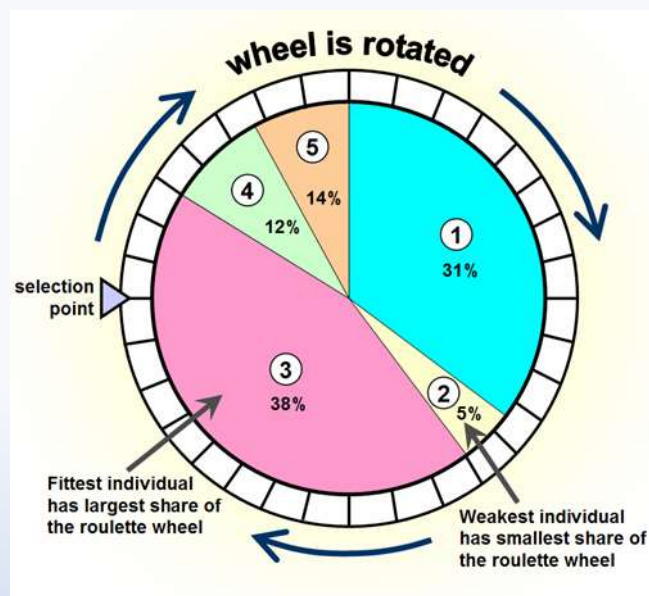
Genetic Algorithm: Deletion



- Generational GA: entire populations replaced with each iteration
- Steady-state GA: a few members replaced each generation

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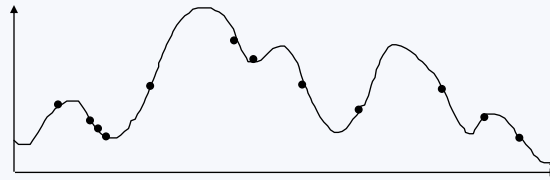
(Roulette Wheel) Selection



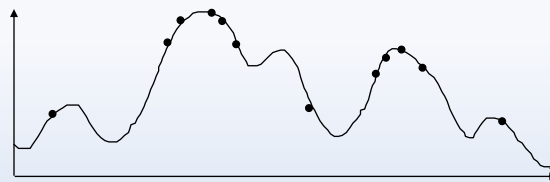
<http://www.edc.ncl.ac.uk/highlight/rhjanuary2007g02.php/>

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An (Abstract) Example



Distribution of Individuals in Generation 0



Distribution of Individuals in Generation N

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A (Less Abstract) Example

The Traveling Salesman Problem

- Find a tour of a given set of cities so that
 - each city is visited only once
 - the total distance traveled is minimized

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Representation

Representation is an ordered list of city numbers known as an order-based GA.

1) London	3) Dunedin	5) Beijing	7) Tokyo
2) Venice	4) Singapore	6) Phoenix	8) Victoria

Population:

- $Tour_1 = (3, 5, 7, 2, 1, 6, 4, 8)$
- $Tour_2 = (2, 5, 7, 6, 8, 1, 3, 4)$

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Crossover



Crossover combines inversion and recombination:

- $Parent_1 = (3, 5, \text{7, 2, 1, 6}, 4, 8)$
- $Parent_2 = (\text{2, 5}, 7, 6, 8, 1, \text{3, 4})$
- $Child = (\text{2, 5}, \text{7, 2, 1, 6}, \text{3, 4})$

Will the crossover operator always generate a valid child that is better (i.e., more fit) than the parents?

- The child will **always be valid** and **always more fit** than the parents
- The child will **always be valid** and **not always more fit** than the parents
- The child will **not always be valid** and **always more fit** than the parents
- The child will **not always be valid** and **not always more fit** than the parents

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Mutation



Mutation involves (in this case) reordering the list of cities:

• Before = (5, 8, 7, 2, 1, 6, 3, 4)

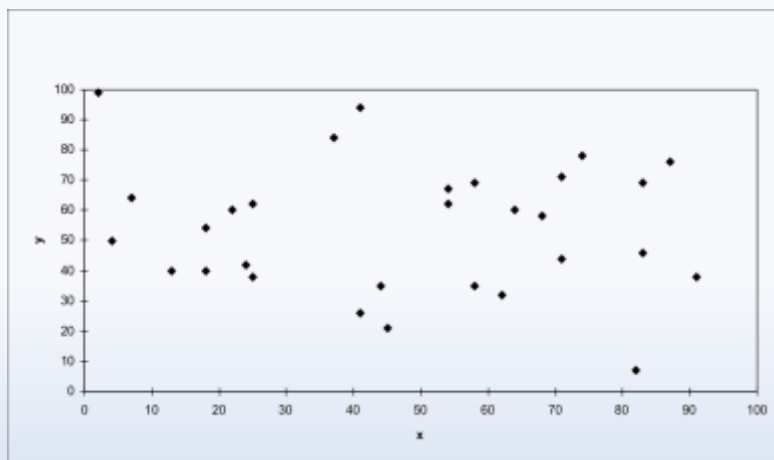
• After = (5, 8, 6, 2, 1, 7, 3, 4)

Will this mutation operator always generate a valid sequence that is better (i.e., more fit) than the unmutated version?

- A. The sequence will **always be valid** and **always more fit** than before
- B. The sequence will **always be valid** and **not always more fit** than before
- C. The sequence will **not always be valid** and **always more fit** than before
- D. The sequence will **not always be valid** and **not always more fit** than before

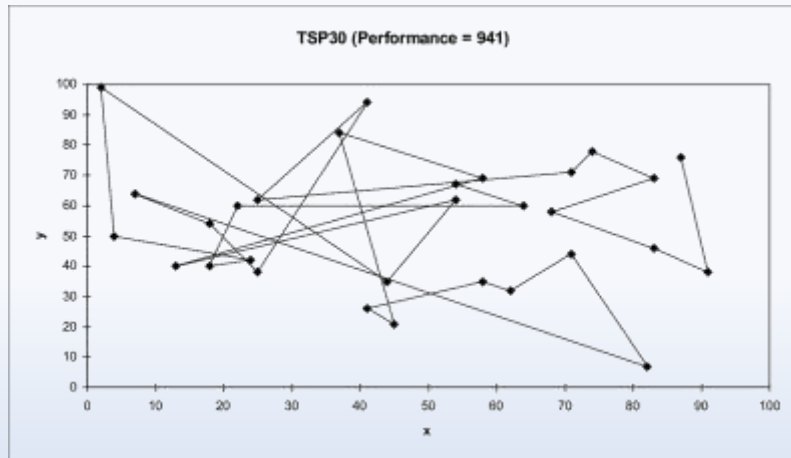
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Traveling Salesman Problem: 30 Cities



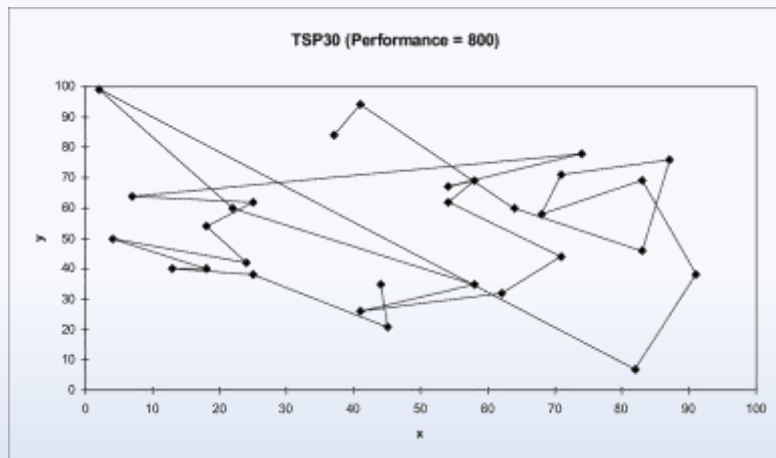
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Early Solution: (Distance = 941)



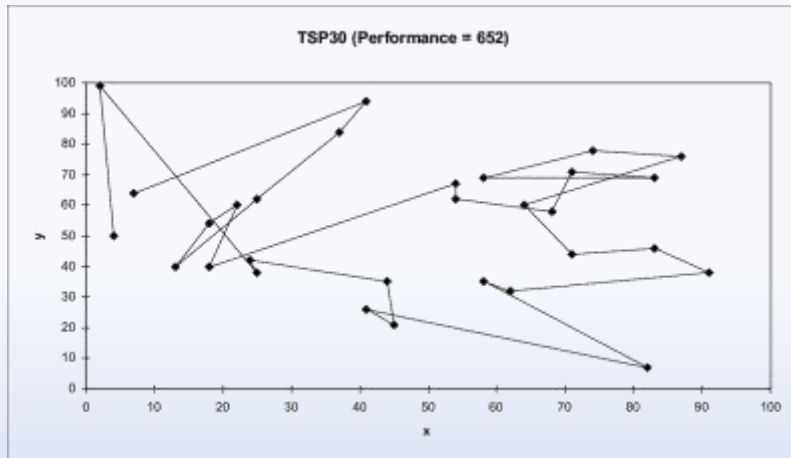
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Later Solution: (Distance = 800)



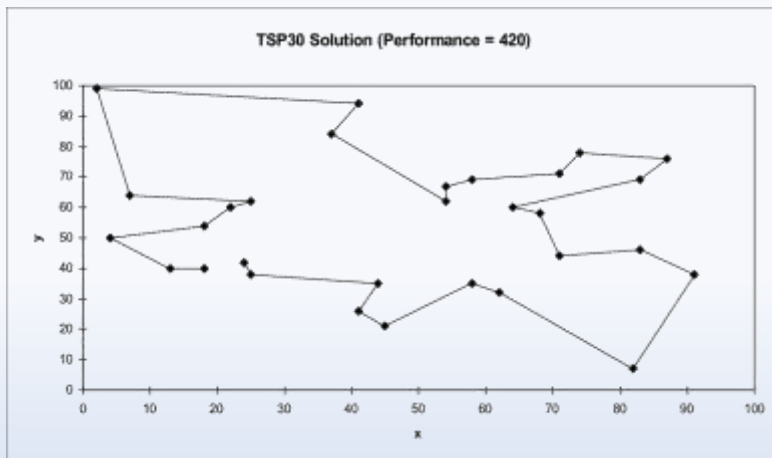
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Much Later Solution: (Distance = 652)



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Best Solution Found: (Distance = 420)



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Optimal Solutions in GAs

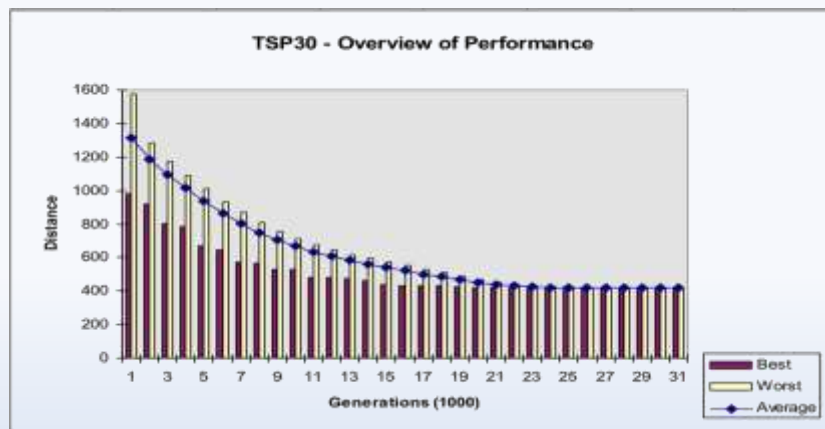


Is the best solution found for the Traveling Salesman Problem optimal (i.e., the best possible solution)?

- A. Yes**, it is the best solution found by the genetic algorithm!
- B. No**, the genetic algorithm might not have found the best solution!

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Rate of Improvement



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Issues for Genetic Algorithms

- Choosing basic implementation issues:
 - representation
 - population size, mutation rate, ...
 - selection, deletion policies
 - crossover, mutation operators
- Termination Criteria
- Performance, scalability
- Solution is only as good as the evaluation function (often hardest part)

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In Class Discussion

Identify a **genetic representation** and **fitness function** for tests in a genetic algorithm.

Possible examples for genetic representation include:

- Bit strings: 0101 ... 1100
- Real numbers: 43.2 -33.1 ... 0.0 89.2)
- Permutations of element: E11 E3 E7 ... E1 E15
- Lists of rules: R1 R2 R3 ... R22 R23
- Program elements: genetic programming
- Or *any* other data structure

Have you represented and identified a fitness function for test case ***inputs*** and ***outputs***?

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Genetic Algorithms

- Bad Problems:
 - Finding Large Primes: Discontinuous and Distributed Solutions
 - A prime is *right* next to something that is not a prime.
 - Being close to a smaller prime doesn't mean you are moving close to a larger prime
 - 2D Path Finding: Search space is small
 - When Random Search = Genetic Algorithm

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Genetic Algorithms

- Pros
 - Faster (and lower memory requirements) than searching a very large search space.
 - Easy, in that if your candidate representation and fitness function are correct, a solution can be found without any explicit analytical work.
 - Good for "noisy" or discrete environments
 - Always an answer; answer gets better with time
- Cons
 - Randomized – not optimal or even complete.
 - Can get stuck on local maxima, though crossover can help mitigate this.
 - It can be hard to work out how best to represent a candidate as a bit string (or otherwise).

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What does this have to do with testing again?

Automatically Generating Tests!

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EvoSuite

- Automatic Test Suite Generation for Java
- Main Features:
 - Generation of JUnit tests for the selected classes
 - Optimization of different coverage criteria, like lines, branches, outputs and mutation testing
 - Tests are minimized: only the ones contributing to achieve coverage are retained
 - Generation of JUnit asserts to capture the current behavior of the tested classes
 - Tests run in a sandbox to prevent potentially dangerous operations
 - Virtual file system
 - Virtual network

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Pynguin

- Automatic Test Suite Generation for Python
- Main Features:
 - Generation of unit tests for the selected modules
 - Tests ***do not*** run in a sandbox to prevent potentially dangerous operations

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Pynguin

<https://www.pynguin.eu/>

Demo

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What now?

Homework!!!

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What now?

<https://swtesting.techconf.org/events/qrs2026>



Home Major Organizers Mootest.net Series [QRS 2026]

The 2026 Summer International Contest on Software Testing at QRS 2026

Software plays an integral role in everyday life due to its near-ubiquitous presence in our increasingly technological society. It has a profound impact on

Important Dates
March 15 - July 22, 2026 Tutorial & Practice Session
July 22, 2026 Registration Deadline
July 23, 2026 (1:00 pm - 5:00 pm, CET (Central European Time)) Contest

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